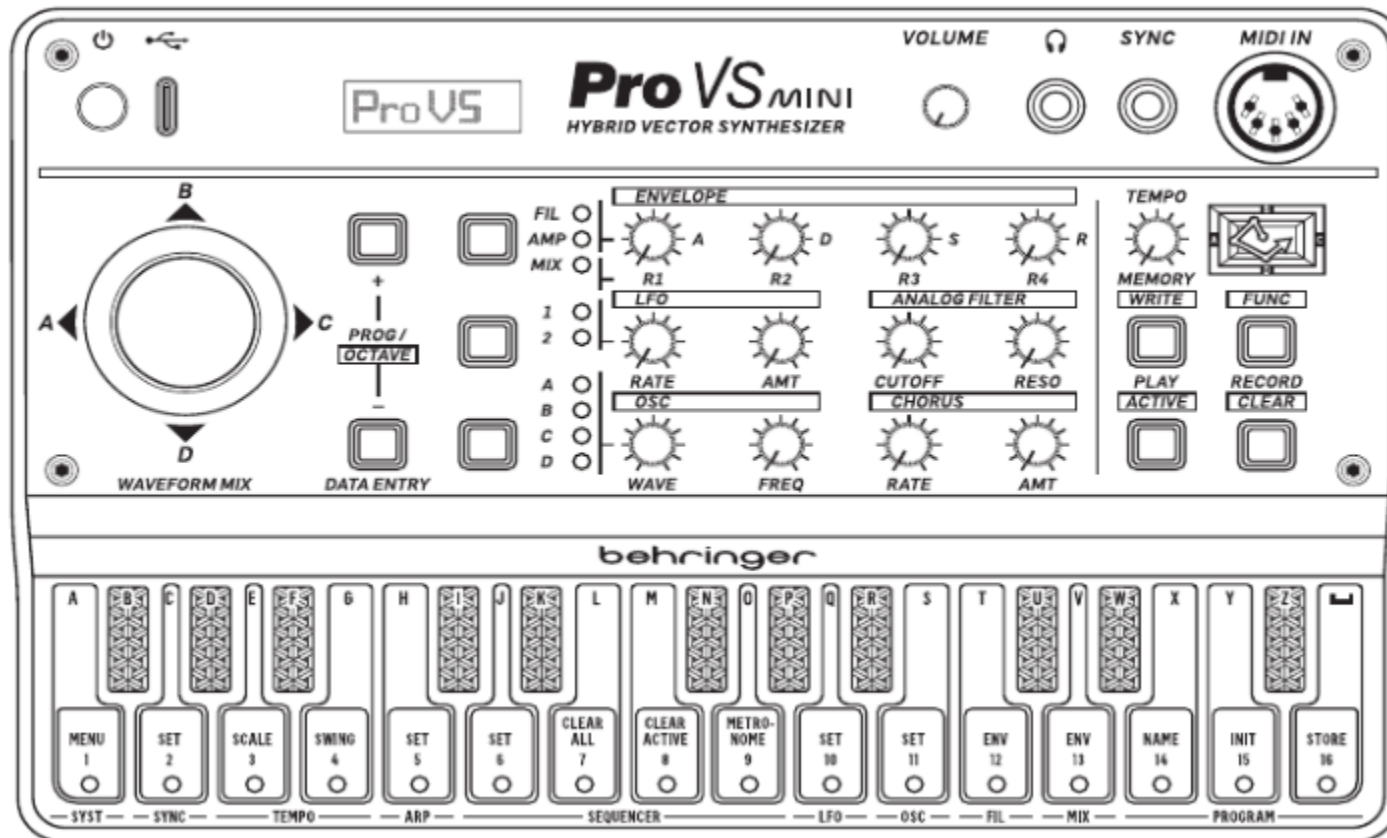


An “Expanded Manual” for the Behringer Pro VS Mini Software version 2.0.2

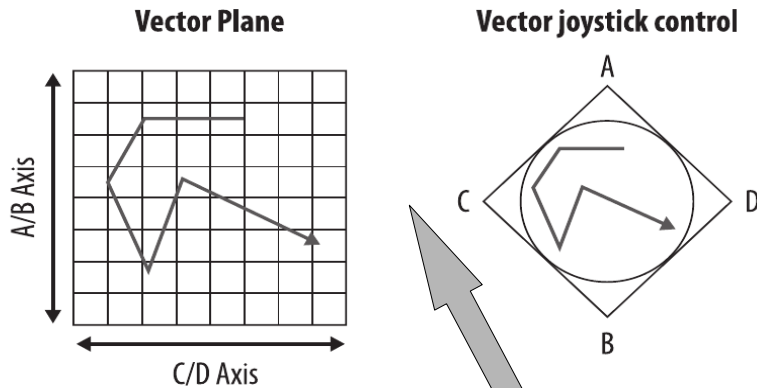
Taken from a variety of internet sources, this manual attempts fill in some of the information missing from the manual.



EN How Vector Synthesis works

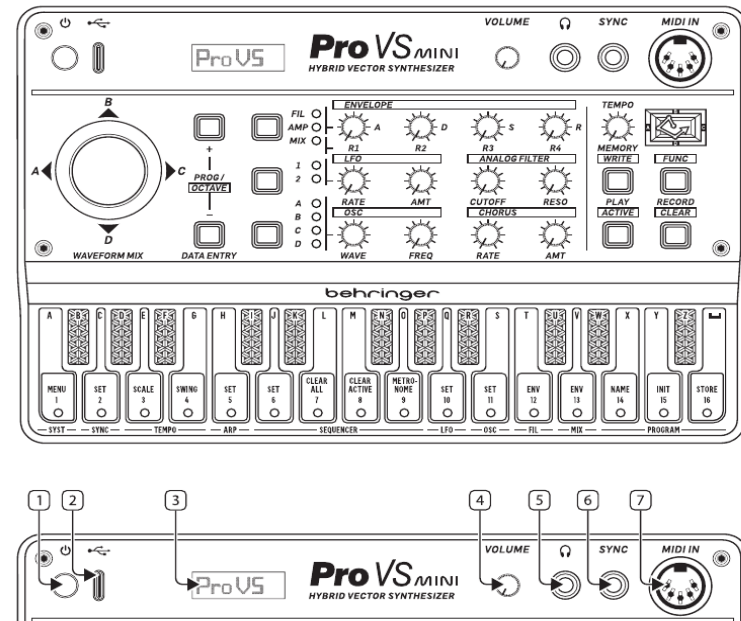
Unlike conventional synthesis, where oscillators are blended together to produce the basis for a sound, Vector Synthesis allows the balance of four oscillators to be continuously adjusted by both the joystick and using the mix envelope. The timbre of the sound can thus be changed subtly or dramatically as you play, depending on the waveforms you pick for each oscillator and how much mix automation you use.

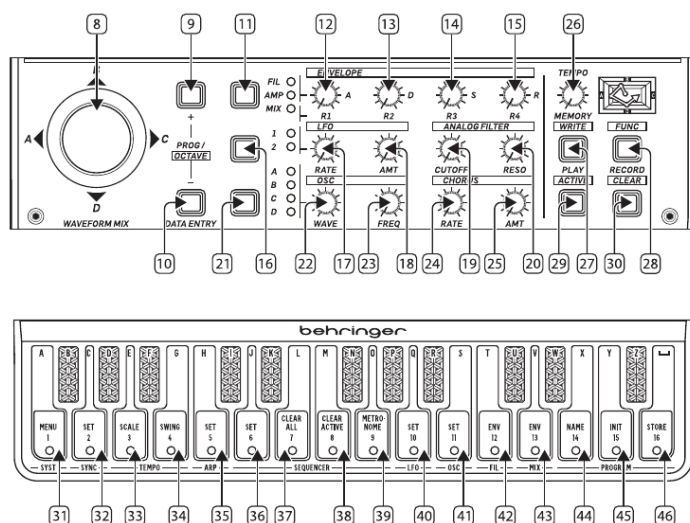
When the waveform mix joystick is in its central position all four oscillators have equal precedence. Moving the joystick upwards will increase the level of oscillator B while decreasing that of oscillator D; A and C remain the same. Equally moving it to the left increases the level of oscillator A against C while not affecting B and D. Therefore, for example, placing the joystick between A and B (North West) would give an equal amount of each of these oscillators while C and D will not contribute. Moving laterally to South East decreases A and B while increasing C and D proportionally, until A and B are not heard.



The result is a sound that morphs within a 2D plane with the joystick movement, mixed with the automation of the mix envelope.

THESE DRAWINGS ARE INCORRECT. The axes are A/C and B/D

PRO VS MINI Controls**EN Step 2: Controls**



PRO VS MINI Controls

EN Step 2: Controls

- POWER** – On/Off button. Hold for 3 seconds to turn the PRO VS MINI on and off.
- USB-C** – Used for power and to connect to a computer for USB MIDI control.
- LCD DISPLAY** – OLED information display.
- VOLUME** – Adjust the output level from the headphones output. Make sure the volume is turned down before putting on headphones.
- PHONES** – Connect your headphones to this 3.5 mm TRS output. Make sure the volume is turned down before putting on headphones.
- SYNC** – This input allows the use of an external clock and start/stop signal on 3.5 mm TRS jack.
- MIDI IN** – Input MIDI from an external MIDI device such as a sequencer or DAW.
- WAVEFORM MIX** – Mix the signals from the 4 sound sources.
- + PROG/OCTAVE** – Increase the program number or raise the octave up of the selected sound.
- PROG/OCTAVE** – Decrease the program number or raise the octave up of the selected sound.
- ENVELOPE SELECT** – Allows the controls to be used by the MIX, FILTER or AMPLITUDE sections.
- R1/ATTACK** – Controls the attack time of the selected envelope.
- R2/DECAY** – Controls the decay time of the selected envelope.
- R3/SUSTAIN** – Controls the sustain level of the selected envelope.
- R4/RELEASE** – Controls the release time of the selected envelope.
- LFO SELECT** – Edit LFO 1 or 2.
- RATE** – Sets the speed of the LFO.
- AMT** – Controls the depth or amount of modulation.
- CUTOFF** – Adjust the 24 dB low-pass filter's cutoff frequency.
- RESO** – Adjusts the resonance of the filter. This emphasizes the frequencies around the cutoff point.
- WAVEFORM SELECT** – Select which one of the 4 voice waveforms, A,B,C or D is active for voice control or editing.

22. **WAVE** – Choose which waveform is used for the currently selected voice.
23. **FREQ** – Change the pitch or frequency of the waveform.
24. **RATE** – Adjust the speed of the chorus.
25. **AMOUNT** – Adjust the chorus depth.
26. **TEMPO** – Sets the speed of the ARP or sequencer in BPM.
27. **MEMORY/WRITE** – Calls up a new sequence memory or, with Function, writes a sequence to internal memory.
28. **FUNC** – Used to engage shift mode or for second functions.
29. **PLAY/ACTIVE** – Plays the arpeggiator or selected sequencer pattern. Function press to make active, which allows selected steps to be removed from the sequence.
30. **RECORD/CLEAR** – Press to enter record or function press to clear notes or a full pattern.
31. **MENU** – Repeated key press gives the following menus:
 - a. **Seq/Arp** – Switches between normal keyboard and sequencer operation and arpeggiator.
 - b. **Filt Env Trig** – Select either Single or Multi trigger. Single triggers the envelope with the first note played and will not re-trigger until all notes are released; Multi will re-trigger with each new note played.
 - c. **MIDI Tx** – Selects MIDI output channel 1 – 16 or off.
 - d. **MIDI Rx** – Selects MIDI input channel 1 – 16 or off.
 - e. **MIDI Velocity** – Switch velocity On or OFF.
 - f. **MIDI CC** – Selects whether MIDI continuous controller messages are received or off.
 - g. **Prog Bulk SysEx Send** – Outputs program memory contents as SysEx using the USB socket.
 - h. **Prog Sing SysEx Send** – Outputs current program as SysEx using the USB socket.
 - i. **Patt Bulk SysEx Send** – Outputs sequencer memory contents as SysEx using the USB socket.
 - j. **Patt Sing SysEx Send** – Outputs current sequence as SysEx using the USB socket.
 - k. **SysEx Receive Listening** – Allows the PRO VS MINI to receive SysEx from an external source using the USB socket.
 - l. **Factory Reset** – Resets all programs and sequences to factory default. Also displays firmware version. Press the + button (9) or – button (10) to initiate reset.
32. **SYNC** – Repeated key press gives the following menus:
 - a. **Sync Source** – Switch between Internal, MIDI, USB or Trigger.
 - b. **Sync Type** – Switch between 1ppq, 2ppq, 4ppq, 24ppq, 48ppq.
33. **SCALE** – Select $\frac{1}{8}$, $\frac{1}{8}T$, $\frac{1}{16}$, $\frac{1}{16}T$, $\frac{1}{32}$ (global).
34. **SWING** – Select 50 (no swing), 55, 60, 65, 70, 75 (global).
35. **ARP SET** – Select up, down, up and down.
36. **SEQUENCER SET** – Switch between step fixed (note lengths defined by step) or record (notes as recorded).
37. **CLEAR ALL** – Wipes all sequencer memories.
38. **CLEAR ACTIVE** – Clears the action of Active Steps (button 29) and makes all steps active.
39. **METRONOME** – Switches the metronome on or off.
40. **LFO SET** – Selects the waveform for the LFO between triangle, square, sawtooth, ramp or S&H.
41. **OSC SET** – Selects coarse or fine tuning for the oscillators using control 23.
42. **FIL ENV** – Adjusts the amount of envelope modulation of the filter from 0 – 99.
43. **MIX ENV** – Repeated key press gives the following menus:
 - a. **Mix Env Loop** – Select between off, 0->3, 1->3, 2->3, 0<->3, 1<->3, 2<->3 (if off is selected then no other menu options are available).
 - b. **Mix Env Repeat** – Select between 1 – 7 and continuous.
 - c. **Mix Env Points: 0-4** – Use the waveform mix control 8 to set the balance between a, b, c and d at each point. Move from point to point with buttons 9 and 10.
44. **NAME** – Allows patch name to be set using keyboard.
45. **PROGRAM INIT** – Initialises the selected program.

46. **STORE** – Stores the current program.

Using the sequencer

- Press the Function button (28) then Menu (31) and select Seq.
- Select fixed steps or Record using key 36.
- Switch metronome on or off as required with key 39.
- Press the Function button again to exit Function mode.
- Put the sequencer into Play using button 29.
- Select the sequence that you want to write to using button 27 and the keyboard (white notes select patterns 1 – 16).
- Use Function and Clear buttons to remove notes from the sequence. Repeatedly pressing clear will clear all notes.
- Notes can then be recorded using the keyboard in time with the metronome (if used).
- The program can be changed at any time during recording using buttons 9 and 10.

- When you are happy with your sequence press Function and Write. Select the pattern number you wish to write to using the keyboard, if you are not writing to the same pattern. Pressing Write again stores your sequence including the program used.
- Press Play to take sequencer out of Record mode. It is advisable to use Function and key 39 to switch off the metronome before playing back.

Using the arpeggiator

- Press the Function button (28) then Menu (31) and select Arp.
- Press Arp Set (key 35) and use buttons 9 and 10 to select Up, Down or Up and Down.
- Press the Function button again to return to normal use.
- Press Play (29).
- Hold down the keys that you want to use for your arpeggio.

PRO-VS MINI Getting Started

EN

EN Step 3: Getting Started

OVERVIEW

This 'getting started' guide will help you set up your PRO VS MINI and briefly introduce its capabilities.

CONNECTION



To connect the PRO VS MINI to your system, please consult the connection guide earlier in this document.

SOFTWARE SETUP

The PRO VS MINI is a USB Class Compliant MIDI device, and so no driver installation is required. The PRO VS MINI does not require any additional drivers to work with Windows and MacOS.

HARDWARE SETUP

Make all the connections in your system. Keep the PRO VS MINI power turned off when making any connections.

Ensure your sound system is turned down.

Turn on the PRO VS MINI before turning on any power amplifiers and turn it off last. This will help prevent any turn on or turn off "pops or thumps" in your speakers.

WARM UP TIME

We recommend leaving 15 minutes or more time for the PRO VS MINI to warm up before recording or live performance. (Longer if it has been brought in from the cold). This will allow the precision analog circuits time to reach their normal operating temperature and tuned performance.

FIRMWARE UPDATE

Please check the behringer.com website regularly for any updates to the Behringer SYNTHTRIBE app. The app looks for the latest firmware file which can then be downloaded and used to update the PRO VS MINI.

The PRO VS MINI has 128 waveforms available for each voice

Number	Name	
0	Twang	
1	Clav	
2	Xylophone	
3	Slippery Slope	
4	Enharmionic Square	
5	Xylophone 2	
6	Mellow Square	
7	Chaos 1	
8	Chaos 2	
9	Chaos 3	
10	Chaos 4	
11	Bright Square	

Number	Name	
12	Hollow	
13	BP Square	
14	Ski Slope 1	
15	Rasp	
16	Peal	
17	Fuzz	
18	Floss	
19	Excite	
20	Awaken	
21	Energize	
22	BP Twang	
23	Complex Pulse	
24	Mellow Square 2	

Number	Name	
25	Harmonic Square	
26	Thin Ramp	
27	Xylophone 3	
28	Thin Square	
29	Harmonic Ramp	
30	Rouse	
31	Waken	
32	Sine	
33	Sawtooth	
34	Square	
35	Dome 1	
36	Dome 2	
37	Mellow Dome	

Number	Name	
38	Dome 3	
39	Dome 4	
40	Rasp 1	
41	Rasp 2	
42	Rasp 3	
43	Rasp 4	
44	Dome 5	
45	HP Saw	
46	High BP Saw	
47	High BP Square	
48	Vocal	
49	Squeeze box	
50	Pulse	

Number	Name	
51	Limp Saw	
52	Spark Wave 1	
53	Spark Wave 2	
54	Spark Wave 3	
55	Mid Wave 1	
56	Cacophonous Buzz 1	
57	Mid Buzz 1	
58	Dome 6	
59	Soft Dome 3	
60	Dome 7	
61	Spark Wave 4	
62	Spark Wave 5	
63	Soft Dome 4	

Number	Name	
64	Dome 8	
65	Organ	
66	Spark Wave 5	
67	Soft Wave 1	
68	Soft Wave 2	
69	Spark Wave 7	
70	Reed	
71	Soft Wave 3	
72	Soft Wave 4	
73	Saxophone	
74	Soft Wave 5	
75	Trashy Wave 1	
76	Trashy Wave 2	

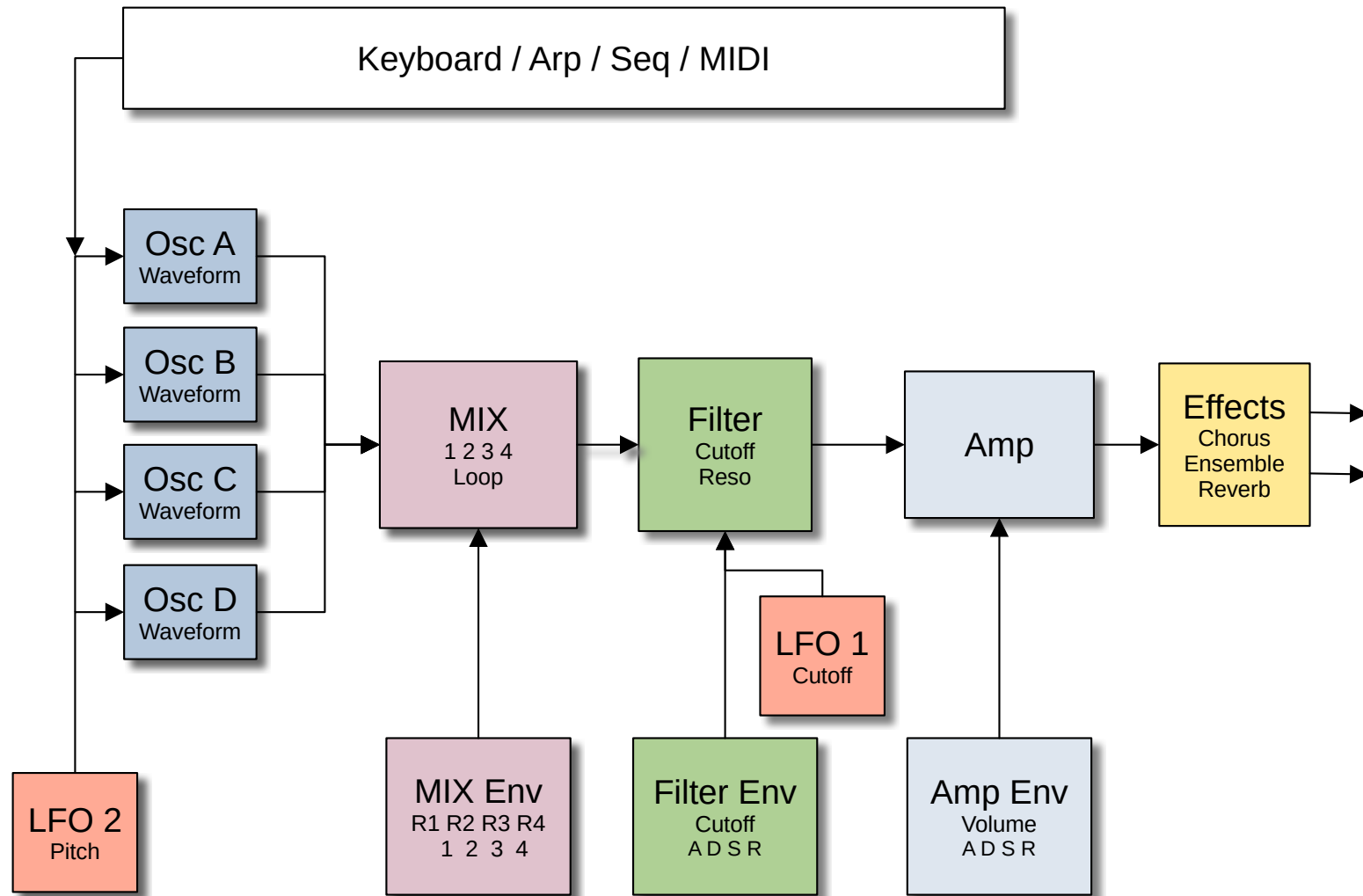
Number	Name	
77	Trashy Wave 3	
78	Trashy Wave 4	
79	Trashy Wave 5	
80	Trashy Wave 6	
81	Jaw Harp 1	
82	Jaw Harp 2	
83	Soft Wave 6	
84	Thin 1	
85	Thin 2	
86	Spark Wave 8	
87	Spark Wave 9	
88	Spark Wave 10	
89	Soft Dome 5	

Number	Name	
90	Soft Dome 6	
91	Cacophonous Buzz 2	
92	Thin 3	
93	Spark Wave 11	
94	Soft Dome 7	
95	Soft Wave 7	
96	Sine 2	
97	Sync'd Sine	
98	Trashy Wave 7	
99	Twingle Pad	
100	Dome 9	
101	Thin 4	
102	Thin 5	

Number	Name	
103	Trashy Wave 8	
104	Trashy Wave 9	
105	Trashy Wave 10	
106	Trashy Wave 11	
107	Trashy Wave 12	
108	Trashy Wave 13	
109	Spark Wave 12	
110	Spark Wave 13	
111	Trashy Wave 14	
112	Trashy Wave 15	
113	Rainbow	
114	Soft Dome 8	
115	Trashy Wave 15	

Number	Name	
116	Soothing 1	
117	Soothing 2	
118	Space Wave	
119	5th Rasp	
120	Octave Wave	
121	Triangle + Overtones	
122	Pulse + Overtones	
123	Rising Square	
124	Soft Wave 8	
125	Cacophonous Buzz 3	
126	Silence	
127	White Noise	

Pro VS mini Block Diagram



This section covers which controls affect which blocks, and how to modify them. I'll use the same button numbers as the Behringer front panel diagrams at the start of this document.

Oscillators A-D	- Encoder 22 selects the wave shape, while pot 23 (Freq) controls the tuning. You can adjust them blindly or press the oscillator select button 21 to display the wave number and tuning offset. Press button 21 again to cycle through all four oscillators. The display will show something like 000 00.00. The left number is the wave number. The right number is the tuning controlled by the Freq pot. - The waves are documented graphically above. Two waves of note are wave 126 – Silence, and wave 127 – White noise. - The amount by which the Freq pot changes the oscillator can be “Coarse” which changes the pitch by a simitone, or “Fine” which changes the pitch by cents. Whether this control is in Coarse or Fine mode is controlled by “Oscillator Set”. - To get to the Osc Set function, press the “Func” button (28) then “Osc Set” (button 41 on the diagram, labeled “11” on the keyboard). Repeatedly pressing the Osc Set button cycles through various oscillator settings. One of them is “Osc Frequency” which sets the function (Coarse or Fine) of the Freq pot. - The second function of Osc Set is Poratmento which controls how quickly or slowly the oscillator can change frequency. This control is sometimes called “glide” on other synthesizers. Zero is the fastest, and 31 is the slowest. - The oscillator pitch is also affected by LFO 2 which will be covered later.
MIX/MIX ENV	The four oscillators are all feeding into a vector mixer. A look at the joystick (control 8) should make it obvious how the mixing works. If you move the joystick while pressing a key, you can control the oscillator mix manually. The real magic of this synth shows up when you automate the motion of the joystick. This automation involves five (5) points numbered 0 through 4. This is MUCH easier if you have an external MIDI keyboard. - To set the Mix Env points, press Func button (28) followed by the “Mix Env” button (labeled 43 on the diagram and 13 on the keyboard). The fist thing the display will show is “Mix Env Loop” which we’ll get to later. Press the Mix Env button again to see “Mix Env Repeat”, and again to see “Mix Env Point: 0”. - The cycle looks like this: When you press a key, the sound starts at Mix Env Point 0, followed by Point 1, 2, and 3. It holds at Point 3 until you release the key. When the key is released it will move to Point 4. The speed at which it changes from one point to the next will be explained later. - Move the joystick to set the starting point of the sound. Usually, this is near oscillator A, but it can be anywhere on the joystick.

	• Press the “+” button (9) to move to Mix Point 1, then Mix Points (2,3,4). • The rate at which the mix moves through the points is controlled by the Mix Envelope. Don’t forget to turn off the Func button. Press button (11) as needed to select the “Mix” envelope. You will see a display showing numbers for “R1 R2 R3 R4”. These are the rates at which the mix moves towards each of the mix points. Use the ADSR pots to adjust each. • Mix Env looping. Use the Func button then Mix Env (43). The first thing you’ll see is the value “0-> 3”. This means that the envelope will loop through Points 0,1,2,3 then back to 0 some number of times (set later) before holding at Point 3 until the key is released. You can use the “+” key to change this setting through various loop cycles. Some of the have a notation like “0<->3”. The loop cycle is 0,1,2,3,2,1,0... until holding at 3 for the release. There is also an “Off” setting. • Pressing the Mix Env button again will take you to “Mix Env Repeat”. This can be set from 1 through 7 loops, as well as continuous looping. • It is important to understand the looping behavior. In a loop like 0->3, it loops through positions 0, 1, 2, 3, 0... Notice the 3 to 0 transition, and understand that there is no “rate” (delay) when the loop recycles, so the jump from 3 back to 0 can cause a significant discontinuity (bump) if these two points are very different. So, if you are going to loop then it makes sense to have the starting point and the point 3 to be pretty close.
FILTER / FILTER ENV	The filter is a simple 24dB low-pass filter. This means that as the Cutoff control (pot 19) is reduced you will hear less of the high frequencies. The Reso[nance] control causes an volume boost of the frequencies near the Cutoff frequency. If you turn this up you get some nice effects, but go too far and the filter will self-oscillate. You may or may not like the result. • The Cutoff frequency can be controlled by the Fil[ter] envelope. Use the envelope select button (11) to select the “Fil” envelope. Use the ADSR pots to control the filter envelope. • A[tack] controls how quickly the filter opens, and D[ecay] controls how quickly it closes to the S[ustain] point. It will remains at the sustain point until you release the key. Then the R[elase] pot controls how quickly the filter closes. • Conceptually, the Filter Envelope opens the filter (turns up the Cutoff frequency) relative to the cutoff pot (19), so if pot 19 is turned way up, then the envelope cannot have any effect. If you turned the cutoff frequency down and still don’t hear anything changing when you press a key, then the Filter Envelope Amount is set too low: The Filter Envelope Amount is controlled by Func “Fil Env” (button 42 or keyboard 12). The amount varies from 0 (no effect) to 99 (very large effect).

AMP / AMP ENV	This controls the volume contour of the sound. Like the filter envelope, use the envelope select button (11) to select the Amp[litude] envelope, then the ADSR pots to control the volume over time.
EFFECTS	The Pro VS mini has a selection of effects. To control the currently selected effect, use the Rate pot in the Chorus (pot 24) to control effect rate, and the Amt pot (25) to control the effect intensity. When the synth was released it only had a Chorus effect, so the section is named “Chorus”. With the more recent software update, there are additional effects. To select the active effect use “Func” “Menu” (button 31 or keyboard 1). You must press it three times to get to the “FX Engine” page. Use the “+” and “-” buttons to choose from Chorus, Ensemble, and Reverb.
LFO 1/2	The LFOs (Low Frequency Oscillators) can be used to continuously alter the pitch of the oscillators and the cutoff frequency of the filter. LFO 1 affects the filter cutoff frequency, and LFO 2 affects oscillator pitch. - Use the LFO select button (16) to choose LFO 1 or LFO 2. - Use the Rate (17) and Amt (18) pots to control the speed and intensity of the effect. - Each LFO can have different waveforms. You can select between Triangle, Square, Sawtooth, and Ramp. To change the waveform you must first select the LFO, then use “Func” “LFO Set” (button 40 or keyboard 10). Use the +/- buttons to choose the waveform. - It is worth knowing that LFO 2 amount is also *set* by the MIDI Mod Wheel (CC 1).

Mix Envelope Behavior

There are two basic types of Mix Envelope behavior; looping and non-looping. Non-looping has “Mix Enc Loop” off.

Non-Looping: When a key is pressed it begins at point P0 and moves to P1 in R(ate) 1 time, then to P2 in R2 time, and finally to P3 in R3 time. It will then hold at P3 until the key is released. It then moves to P4 in R4 time. That's it.

Looping: All loops begin at P0 and end at P3. However, they can loop back to P0, P1, or P2.

0 → 3 will go through P0, P1, P2, P3, then back to P0, P1, P2, P3, ...

1 → 3 will go through P0, P1, P2, P3, then back to P1, P2, P3, ...

2 → 3 will go through P1, P2, P3, then back to P2, P3, ...

On the following reversing loops, the rates operate in reverse as well.

0 ← → 3 will go through P0, P1, P2, P3, P2, P1, P0, ...

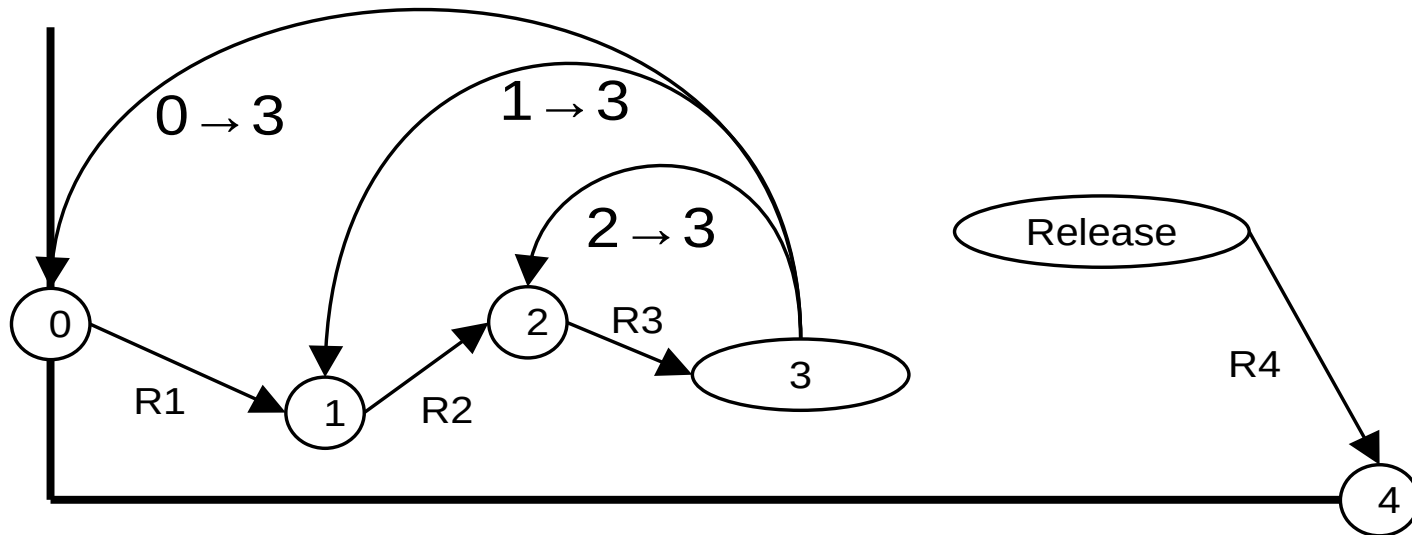
1 ← → 3 will go through P0, P1, P2, P3, P2, P1, P2, P3, P2, P1, ...

2 ← → 3 will go through P0, P1, P2, P3, P2, P3, P2, ...

Loops can be set to repeat a set number of times (1-7) before holding at P3, or to repeat continuously.

Because there is no delay between P3 and returning to the start of the loop (P0, P1, or P2) it will sound better if the end point and the loop start point are close together. Otherwise there will be a noticeable discontinuity.

Regardless of where the loop process is, when the key is released the mix envelope will move from WHEREVER it is to P4 in R4 time.



MIDI CC Implementation from <https://midi.guide/d/behringer/pro-vs-mini/>
and <https://deepwiki.com/gabrielinuz/midi-cc-controller/5.1-behringer-pro-vs-mini>

Notes:

- The Pro VS Mini does not respond to CC7 (Volume)

General

Parameter	CC	Range	Usage	Orientation
Modulation	1	[0–127]	—	0-BASED
Portamento time	5	[0–127]	—	0-BASED

FX

Parameter	CC	Range	Usage	Orientation
FX engine	9	[0–127]	0~42: Chorus; 43~84: Ensemble; 85~127: Reverb	0-BASED
FX rate	91	[0–127]	—	
FX amt	92	[0–127]	—	

Voice wave

Parameter	CC	Range	Usage	Orientation
Voice A	24	[0–127]	See list above	0-BASED
Voice B	25	[0–127]	See list above	0-BASED
Voice C	26	[0–127]	See list above	0-BASED
Voice D	27	[0–127]	See list above	0-BASED

Voice Tuning

Parameter	CC	Range	Usage	Orientation
Voice A	111	[0–127]	Fine Tune (cents)	0-BASED
Voice B	112	[0–127]	Fine Tune (cents)	0-BASED
Voice C	113	[0–127]	Fine Tune (cents)	0-BASED
Voice D	114	[0–127]	Fine Tune (cents)	0-BASED
Voice A	115	[0–127]	Coarse Tune (semitones)	0-BASED
Voice B	116	[0–127]	Coarse Tune (semitones)	0-BASED
Voice C	117	[0–127]	Coarse Tune (semitones)	0-BASED
Voice D	118	[0–127]	Coarse Tune (semitones)	0-BASED

Filter

Parameter	CC	Range	Usage	Orientation
Filter env amount	47	[0–127]	—	0-BASED
Filter resonance	71	[0–127]	—	0-BASED
Filter Cutoff	74	[0–127]	—	0-BASED
Env A	85	[0–127]	—	0-BASED
Env D	86	[0–127]	—	0-BASED
Env S	87	[0–127]	—	0-BASED
Env R	88	[0–127]	—	0-BASED

Amp

Parameter	CC	Range	Usage	Orientation
Env A	81	[0–127]	—	0-BASED
Env D	82	[0–127]	—	0-BASED
Env S	83	[0–127]	—	0-BASED
Env R	84	[0–127]	—	0-BASED

LFO

Parameter	CC	Range	Usage	Description	Orientation
LFO 1 waveform	54	[0–127]	—	Triangle, square, saw, ramp, s&h	0-BASED
LFO 1 amount	70	[0–127]	—	—	0-BASED
LFO 1 rate	72	[0–127]	—	—	0-BASED
LFO 2 waveform	55	[0–127]	—	Triangle, square, saw, ramp, s&h	0-BASED
LFO 2 amount	28	[0–127]	—	—	0-BASED
LFO 2 rate	73	[0–127]	—	—	